Flow CRM Data Quality Validation Report

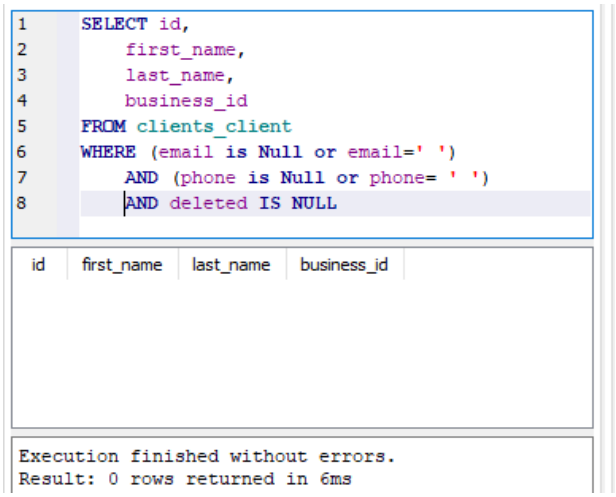
VALIDATION METADATA
Date: February 13, 2026
Tested by: Data Analyst Alex Kvasnevska
Database: flow_crm_clients
Standards Reference: [FlowCRM Data Quality Standards Version 1.0]

QUERY RESULTS:

Query #1: Clients without contact info
Purpose: Find clients missing both email AND phone
Expected: 0 records

```
SELECT id,
        first_name,
        last_name,
        business_id
FROM clients_client
WHERE (email IS NULL OR email = '')
        AND (phone IS NULL OR phone = '')
        AND deleted IS NULL
```

Screenshot:



Result: 0 records
Status: PASSED

Query #2: Duplicate e-mails
Purpose: detect duplicates emails without alight client
Expected: 0 records

```
SELECT  email,
        business_id,
        count(*)
FROM clients_client
WHERE  deleted IS Null
Group by 1,2
HAVING count(*)>1
```

```
1   SELECT email,
2       business_id,
3       count(*)
4   FROM clients_client
5   WHERE deleted is NULL
6   GROUP BY 1,2
7   HAVING count(*)>1
```

email	business_id	count(*)

```
Execution finished without errors.
Result: 0 rows returned in 8ms
```

Results: 0 records

Status: PASSED

Query #3 Invalid status values

Purpose: find the client with a status outside the allowed values

Expected: 0 records

```
SELECT id,
        first_name,
        last_name,
        status
FROM clients_client
WHERE status NOT IN ('New', 'Done', 'In Progress')
```

```
1   SELECT id,
2       first_name,
3       last_name,
4       status
5   FROM clients_client
6   WHERE status NOT IN('New', 'Done','In Progress')
```

id	first_name	last_name	status

```
Execution finished without errors.
Result: 0 rows returned in 5ms
```

Results: o rows

Status: PASSED

Query #4: Logical consistency (Date Asynchrony)
Purpose: detect records where the update timestamp is earlier than the creation timestamp
Expected results: 0 records

```
SELECT id,
        first_name,
        last_name,
        updated_at,
        created_at
FROM clients_client
WHERE created_at> updated_at
```

```
1  SELECT id,
2      first_name,
3      last_name,
4      updated_at,
5      created_at
6  FROM clients_client
7  WHERE created_at> updated_at
```

	id	first_name	last_name	updated_at	created_at
1	6	Тарас	Петренко	2026-01-10 07:08:26	2026-01-10 08:20:26
2	16	Robert	Lee	2026-01-21 08:51:26	2026-01-26 12:55:26
3	26	Іван	Іваненко	2026-01-23 07:25:26	2026-01-28 12:21:26
4	29	Максим	Поліщук	2026-01-18 05:38:26	2026-01-31 09:21:26
5	31	Sven	Nielsen	2026-01-18 06:15:26	2026-02-02 12:45:26
6	44	Назар	Шевченко	2026-01-17 08:36:26	2026-02-05 05:35:26
7	49	Андрій	Романенко	2026-01-14 05:24:26	2026-01-28 13:00:26
8	50	Sven	Johansson	2026-01-14 13:41:26	2026-01-30 05:40:26
9	61	Barbara	Jones	2026-01-14 04:51:26	2026-01-16 04:01:26
10	82	Per	Olsen	2026-01-28 14:39:26	2026-01-31 13:22:26
11	100	Greta	Johansson	2026-02-01 08:28:26	2026-02-07 11:45:26
12	104	Lisa	Moore	2026-01-17 13:56:26	2026-01-25 10:11:26
13	116	José	García	2026-01-10 09:17:26	2026-02-08 12:08:26

```
Execution finished without errors.
Result: 13 rows returned in 6ms
```

Results: 13 rows
Status: FAILED

Query #5: Data validity (Phone number length)
Purpose: identify phone numbers that are suspiciously short (less than 7 characters) or long (more than 15 characters)
Expected results: 0 records

```
SELECT id,
        first_name,
        last_name,
        length(phone) as phone_length
FROM clients_client
WHERE phone IS NOT NULL
        AND phone!=' '
        AND length(phone)<7
        OR length(phone)>15
```

```
SELECT id,
    first_name,
    last_name,
    length(phone) as phone_length
FROM clients_client
WHERE phone IS NOT NULL
    AND phone!=' '
    AND length(phone)<7
    OR length(phone)>15
```

	id	first_name	last_name	phone_length
1	114			0

```
Execution finished without errors.
Result: 1 rows returned in 5ms
```

Results: 1 rows
Status: FAILED

Query #6: Content quality (empty or extra-long notes)
Purpose: indicate comment which were saved empty or with a length over 300 characters
Expected results: 0 records

```
SELECT id, first_name, last_name, comment
FROM clients_client
WHERE (comment IS NULL OR comment = '')
  OR length(comment) >= 300
```

```
SELECT id, first_name, last_name, comment
FROM clients_client
WHERE (comment IS NULL OR comment = '')
   OR length(comment) >= 300
```

	id	first_name	last_name	comment
1	1	Betty	Miller	
2	43	Андрій	Семенов	
3	52	Joseph	Miller	
4	61	Barbara	Jones	
5	68	Henrik	Eriksson	
6	95	Ганна	Петренко	
7	101	Оксана	Романенко	
8	117	Testing	VeryLongCommentField	This is an extremely long comment …

```
Execution finished without errors.
Result: 8 rows returned in 4ms
```

Results: 8 rows

Status: FAILED

Query #7: Referential integrity (orphaned clients)

Purpose: ensure every client is linked to en existent business in the business_business table

Expected results: 0 records

```
SELECT c.id,
        c.first_name,
        c.last_name,
        c.business_id
FROM clients_client as c
LEFT JOIN business_business as b
ON c.business_id= b.id
WHERE b.id IS NULL
```

```sql
SELECT c.id,
    c.first_name,
    c.last_name,
    c.business_id
FROM clients_client as c
LEFT JOIN business_business as b
ON c.business_id= b.id
WHERE b.id IS NULL
```

	id	first_name	last_name	business_id
1	1	Betty	Miller	1
2	2	Astrid	Pettersson	1
3	3	Ірина	Марченко	1
4	4	Barbara	Garcia	1
5	5	Linda	Martinez	1
6	6	Тарас	Петренко	1
7	7	Patricia	Lopez	1
8	8	Артем	Павленко	1
9	9	James	Wilson	1

```
Execution finished without errors.
Result: 120 rows returned in 6ms
```

Status: FAILED / ESCALATED TO BACKEND

Observation: Despite attempts to manually remediate the data, the system blocked updates with multiple schema-level errors (NOT NULL and FOREIGN KEY constraints).

Root Cause: The database architecture lacks default values for mandatory system fields, and existing constraints prevent the necessary relational hierarchy from being established manually.

Conclusion: Data integrity cannot be restored through manual seeding. A backend fix is required to align the CSV import process with database constraints.

SQL 1*

```sql
-- 1. Create minimal User
INSERT OR IGNORE INTO users_user (id, password, first_name, last_name, email, is_active)
VALUES (1, 'password123', 'System', 'Admin', 'admin@flowcrm.com', 1);

-- 2. Create Business
INSERT OR IGNORE INTO business_business (id, name, type, email, phone, description, created_at, updated_at, owner_id)
VALUES (1, 'Default Business', 'Service', 'admin@flowcrm.com',
'+380000000000', 'Validation record', '2026-02-13 12:00:00', '2026-02-13 12:00:00', 1);
```

DB Browser for SQLite

Помилка під час збереження файлу бази даних. Це означає, що не всі зміни в базу даних було збережено. Спочатку Вам необхідно розв'язати таку помилку.

FOREIGN KEY constraint failed (RELEASE "UNDOPOINT";)

OK

```
Execution finished without errors.
Result: query executed successfully. Took 0ms
At line 5:
-- 2. Create Business
INSERT OR IGNORE INTO business_business (id, name, type, email, phone, description, created_at, updated_at, owner_id)
VALUES (1, 'Default Business', 'Service', 'admin@flowcrm.com',
'+380000000000', 'Validation record', '2026-02-13 12:00:00', '2026-02-13 12:00:00', 1);
```

Query #8: Dummy data detection
Purpose: recognize test records that should not be in production
Expected results: 0 records

```
SELECT id,
        first_name,
        last_name,
        email
FROM clients_client
Where first_name LIKE '%test%'
        OR last_name LIKE '%test%'
        OR email LIKE '%test%'
```

```
1   SELECT id,
2       first_name,
3       last_name,
4       email
5   FROM clients_client
6   Where first_name LIKE '%test%'
7       OR last_name LIKE '%test%'
8       OR email LIKE '%test%'
```

	id	first_name	last_name	email
1	114			noreply@testcompany.com
2	117	Testing	VeryLongCommentField	test@example.com

```
Execution finished without errors.
Result: 2 rows returned in 4ms
```

Results: 2 rows
Status: FAILED

Query #9: The multiply identity email
Purpose: detect cases where the same email is assigned to different clients
Expected results: 0 records

```
SELECT email,
    COUNT(DISTINCT first_name) as unique_names_count,
    GROUP_CONCAT(DISTINCT first_name) as names_found
FROM clients_client
WHERE email IS NOT NULL AND email != ''
GROUP BY email
HAVING unique_names_count > 1
```

```sql
SELECT email,
       COUNT(DISTINCT first_name) as unique_names_count,
       GROUP_CONCAT(DISTINCT first_name) as names_found
FROM clients_client
WHERE email IS NOT NULL AND email != ''
GROUP BY email
HAVING unique_names_count > 1
```

email	unique_names_count	names_found

```
Execution finished without errors.
Result: 0 rows returned in 3ms
```

Results: 0 rows

Status: PASSED

Query #10: Suspicious email lengths

Purpose: find a technically correct but physically improbable email

Expected results: 0 records

```sql
SELECT id,
        email,
        length(email) as email_length
From clients_client
Where length(email)<6
        OR length(email)>50
        OR email NOT LIKE '%_@_%._%'
```

```sql
SELECT id,
    email,
    length(email) as email_length
From clients_client
Where length(email)<6
    OR length(email)>50
    OR email NOT LIKE '%_@_%._%'
```

id	email	email_length

```
Execution finished without errors.
Result: 0 rows returned in 4ms
```

Results: 0 rows

Status: PASSED

SUMMARY:
Passed: 5/10
Failed: 5/10

Audit Score: 50% Success Rate (5/10 Checks Passed).
Key Finding: While the database maintains basic uniqueness and email formatting, it significantly fails in Structural Integrity and Temporal Logic.
Critical Issue: The presence of 120 "orphaned" records (Query #7) combined with rigid database constraints makes manual cleanup impossible.
This indicates a high risk of data loss or corruption during future migrations.
Data Hygiene: Minor failed checks in phone lengths and dummy data (test records) suggest a lack of front-end validation.

ACTION ITEMS:

1. Immediate Backend Intervention (Priority: Critical):
    - Resolve the Referential Integrity failure. All 120 orphaned clients must be mapped to a valid `business_id` or investigated for deletion.
    - Review database constraints for the `business_business` table. Missing default values for `created_at` and `owner_id` are currently blocking administrative data recovery.
2. Logic Fix (Priority: High):
    - Investigate the 13 records where `created_at > updated_at`. This is a logical paradox that suggests a bug in the system clock or the backend's timestamping logic.
3. Front-End Validation Update (Priority: Medium):
    - Implement stricter masks for Phone Numbers (min 7, max 15 chars) to eliminate invalid entries found in Query #5.
    - Add a "Trim" function for Comments and a character limit (max 300) to prevent UI breakage and storage bloat.
4. Database Cleanup (Priority: Low):
    - Purge test/dummy records identified in Query #8 before moving to the production environment.